GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST 2

Academic Period : January 2026

Subject	:	**WIRELESS NETWORK & SECURITY**
Code	:	**CBS 2343**
Duration	:	**3 HOURS**
Total Marks	:	**50 MARKS**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Read the specific instructions and questions carefully before answering.

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Semester/Trade/Group	:	SEM 4 / DCBS / CBS 1
Date	:	7/5/2026

Code	CBS 2343	Course Name	WIRELESS NETWORK SECURITY
Chapter	Chapter 5: WLAN security (management frame)	Examiner	
Name		Program/ Group	DNWS/

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1	To capture Wireshark file	2	2	P3	2			
2	To identify beacon frame	2	2	P3	15			
3	To identify authentication frame	2	2	P3	12			
4	To identify de-authentication frame	2	2	P3	9			
5	To identify wireless attack	2	2	P3	12			
				Total	**50**			
				Total Marks				
			PO3	/50				
		Total						
			Signature					

PRACTICAL INFORMATION

Three types of 802.11 frames:
1. Management frame
2. Control frame
3. Data frame

Management frames are used to manage the BSS, control frames access to the medium, and data frames contain payloads that are the layer 3 – 7 information.

This practical will focus on the contents of management frames.

Students need to install Wireshark the latest version. All answers are based on the Wireshark file provided by TTO/Lecture.

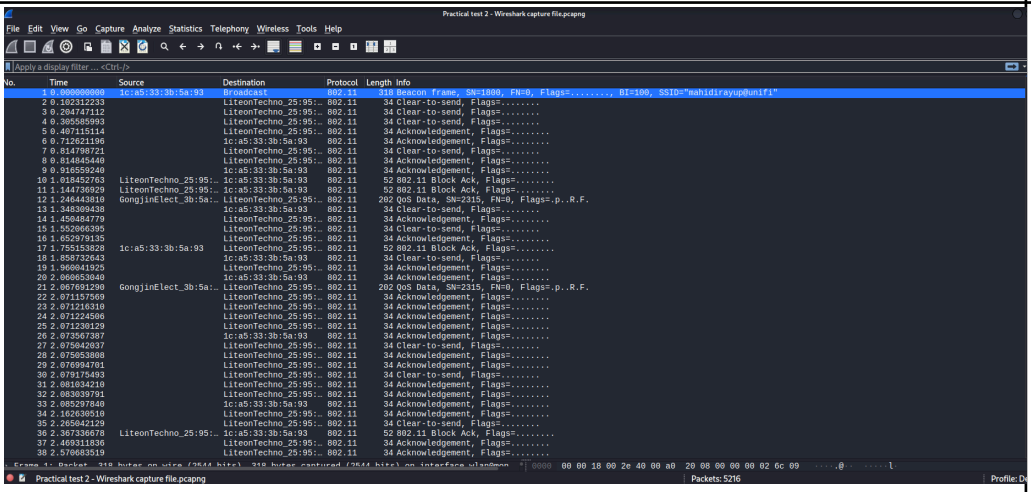
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Learning Objectives

Part 1: To identify MAC addressing
Part 2: To identify Beacon frame
Part 3: To identify Authentication frame
Part 4: To identify De-Authentication frame
Part 5: To identify wireless attack

Answer all questions.

All questions are based on your Wireshark capture file .pcapng **provided by TTO**.

Wireshark file:	Include screenshots of the packet capture results.	Marks
Provided by TTO		**2**

In Wireshark, to filter	Questions, Verification, & Answer: (List the evidence and include screenshots of the results.)	Marks
wlan.fc.type_subtype eq 8	What is MAC address for wireless router? Source address: 1c:a5:33:3b:5a:93 (1c:a5:33:3b:5a:93)	**3**
	What is MAC address for Client? Destination address: Broadcast (ff:ff:ff:ff:ff:ff)	**3**
	What is SSID name? SSID="mahidirayup@unifi"	**3**
	What is wireless standard used by client? 802.11 radio information PHY type: 802.11b (HR/DSSS) (4)	**3**
	What is frame Control Field for beacon frame? Frame Control Field: 0x8000 .... ..00 = Version: 0 .... 00.. = Type: Management frame (0) 1000 .... = Subtype: 8	**3**

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wlan.fc.type_subtype eq 11	What is MAC address for Client? ▾ Destination address: 1c:a5:33:3b:5a:93 (1c:a5:33:3b:5a:93)	**3**
	What is MAC address for wireless router? ▸ Source address: LiteonTechno_25:95:0f (48:d2:24:25:95:0f)	**3**
	What is frame control field for authentication frame? Type/Subtype: Authentication (0x000b) ▾ Frame Control Field: 0xb000 .... ..00 = Version: 0 .... 00.. = Type: Management frame (0) 1011 .... = Subtype: 11 ▸ Flags: 0x00 .000 0001 0011 1010 = Duration: 314 microseconds	**3**
	How many clients are connected to the wireless router? LiteonTechno_25:95:… 802.11 ChinaMobileG_41:8d:… 802.11 TpLinkTechno_1f:16:… 802.11	**3**
wlan.fc.type_subtype eq 12	What is MAC address for client? LiteonTechno_25:95:0f (48:d2:24:25:95:0f) 48:d2:24:25:95:0f	**3**
	What is the MAC address of the wireless router that is connected to the client? * Hint: It is related to question No. 1. Source address: 1c:a5:33:3b:5a:93 (1c:a5:33:3b:5a:93)	**3**
	What is frame Control Field for de-authentication frame? ▾ IEEE 802.11 Deauthentication, Flags: .... Type/Subtype: Deauthentication (0x000c) ▾ Frame Control Field: 0xc000 .... ..00 = Version: 0 .... 00.. = Type: Management frame (0) 1100 .... = Subtype: 12	**3**
To investigate a wireless attack: Students can filter frames by subtype number to identify the attacker.	What is MAC address attacker? 48:d2:24:25:95:0f	**4**
	What is the manufacturer or model being used by the attacker to attack the wireless router? Liteon Technology Corporation Aug 27, 2015 4F,90,Chien 1 Road,ChungHo New Taipei City Taipei TW 23585 Detected Changes: Address 2015	**4**

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	What is the subtype frame being used by the attacker to attack the wireless router? ▾ IEEE 802.11 Deauthentication, Flags: .... Type/Subtype: Deauthentication (0x000c)	**4**

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